

Why Trust the Sunrise? Hume and the Problem of Induction

An original lesson plan inspired by **Leah Henderson's** the *Stanford Encyclopedia of Philosophy* entry on *The Problem of Induction*. Not affiliated with or endorsed by *Stanford Encyclopedia of Philosophy* or Leah Henderson. Explore the original course: <https://plato.stanford.edu/entries/induction-problem/>

GRADE LEVEL

Grades 11-12 / Intro College

DURATION

60 minutes

TOPICS

Epistemology

“You believe the sun will rise tomorrow because it always has before — but can that kind of reasoning ever be rationally justified without secretly assuming the very thing it's trying to prove?”

Learning Objectives

By the end of this lesson, students will be able to:

- Explain Hume's argument that inductive inferences can be justified neither demonstratively nor by experience without circularity.
- Apply Hume's argument structure to a concrete case of inductive reasoning to identify exactly where an unproven uniformity assumption is required.
- Evaluate a proposed response to Hume's problem — Bayesian reframing or Reichenbach's pragmatic vindication — and defend a judgment about whether it resolves the problem or merely reframes it.

Materials Needed

- Handout with Hume's original bread/nourishment example and the demonstrative-vs-probable distinction
- A bag of colored marbles or tokens for a live sampling demonstration
- Whiteboard for diagramming the circularity of using induction to justify induction
- Index cards for closing position statements

Lesson Procedure

Hook: The Marble Bag

7 min

Draw ten marbles from an opaque bag, all red. Ask students how confident they are that the eleventh marble will also be red, and why. Press on what exactly justifies extending the observed pattern to an unobserved marble.

Direct Instruction: Hume's Argument

18 min

Present Hume's distinction between "relations of ideas" (demonstrative, a priori truths like mathematics, whose denial is a contradiction) and "matters of fact" (based on experience, whose denial is never a contradiction). Show that inductive inferences from past to future rest on an assumption that "the future will resemble the past," and that this assumption can't be proven demonstratively (its denial isn't a contradiction) or by experience (since using induction's past success to justify induction is circular, assuming the very uniformity principle in question).

Small-Group Activity: Locating the Gap

18 min

Groups work through Hume's bread example — bread has nourished me in the past, therefore it will nourish me in the future — identifying precisely where an unstated uniformity assumption is smuggled in, then attempt to patch the argument themselves and check whether their patch also secretly relies on an unproven uniformity assumption.

Direct Instruction and Discussion: Responses to Hume

12 min

Briefly present two responses: a Bayesian/probabilistic reframing (induction doesn't need certainty, just rational degrees of belief updated by evidence, though justifying the priors and updating rule raises similar worries) and Reichenbach's pragmatic vindication (we can't prove induction works, but if any method could predict the future, induction would, so using it is our best bet even without a guarantee). Discuss whether either actually answers Hume or simply lowers the bar for what counts as an answer.

Closure

5 min

Index-card verdict: does the problem of induction have a real solution, or do we just have no choice but to keep using induction anyway?

Discussion Questions

- 1 Is "the future will resemble the past" something we could ever prove without already assuming it, or relying on the very kind of reasoning that's in question?
- 2 Does the demonstrative/empirical distinction Hume draws really exhaust the possibilities, or might there be a third kind of justification that escapes his dilemma?
- 3 Does Reichenbach's pragmatic vindication — "if anything works, induction will" — actually answer Hume's challenge, or does it concede that induction can't be justified and settle for a fallback strategy instead?
- 4 If a Bayesian says we just need reasonable "priors" and updating rules rather than certainty, doesn't the question of what makes a prior reasonable raise a version of the same problem all over again?
- 5 If Hume is right that induction can't be rationally justified, does that mean scientific reasoning itself is irrational, or does science not actually depend on solving this philosophical puzzle to keep functioning?

Assessment

Students write a short paper applying Hume's argument structure to a real inductive inference they rely on in daily life (for example, "this bridge has held cars before, so it will hold my car"), identifying precisely where the uniformity



assumption is smuggled in, and evaluating whether the Bayesian or Reichenbach-style response actually resolves the gap. Grade on accurate reconstruction of Hume's argument and genuine engagement with a response to it.

Extension Activity

Have students research how working scientists or statisticians talk about the reliability of inductive or statistical inference — such as confidence intervals or the replication crisis — and write a short memo on whether practicing scientists seem to treat Hume's problem as a live concern or as a solved, irrelevant issue.

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